



3 -Month DSA Roadmap (Beginner → Interview Ready)



Goal

Become **interview-ready** in Data Structures & Algorithms **within 3 months** by building strong fundamentals, problem-solving ability, and coding confidence.



Overall Structure

Phase	Duration		Focus
Phase 1	Week	1 – 4	Arrays + Strings (Core Logic Building)
Phase 2	Week	5 – 8	Linear Data Structures
Phase 3	Week	9 – 12	Non-Linear DS + Interview Prep



Phase 1 — Logic Foundation (Week 1 – 4)

Topics

- Arrays basics
- Two pointers
- Sliding window
- HashMap & HashSet
- Basic math logic
- String manipulation

Daily Routine (1 . 5 – 2 hrs)

- 1 . Revise concept — 20 mins
- 2 . Solve 2 Easy problems — 60 mins
- 3 . Review solutions — 20 mins

Problems (LeetCode Numbers)

- 1920 — Build Array from Permutation
- 136 — Single Number
- 26 — Remove Duplicates from Sorted Array

- 2 8 3 — Move Zeroes
- 1 6 7 2 — Richest Customer Wealth
- 1 2 1 — Best Time to Buy and Sell Stock
- 2 1 7 — Contains Duplicate
- 3 4 4 — Reverse String
- 1 2 5 — Valid Palindrome

Target by Week 4

✓ Think independently ✓ Write code without copying ✓ Understand time complexity

✓ Phase 2 — Linear Data Structures (Week 5 – 8)

Topics

- Linked List
- Stack
- Queue
- Recursion basics

Week Plan

Week 5 — Linked List

- Traversal
- Insert/Delete
- Reverse Linked List

Problems: - 2 0 6 — Reverse Linked List - 2 1 — Merge Two Sorted Lists - 1 4 1 — Linked List Cycle

Week 6 — Stack

- Stack using array
- Valid parentheses
- Monotonic stack idea

Problems: - 2 0 — Valid Parentheses - 1 5 5 — Min Stack - 4 9 6 — Next Greater Element I

Week 7 — Queue

- Queue basics
- Circular queue
- BFS intuition

Problems: - 9 3 3 — Number of Recent Calls - 2 2 5 — Implement Stack using Queues

Week 8 — Recursion Intro

- Base case thinking
- Recursive patterns

Problems: - 5 0 9 — Fibonacci Number - 7 0 — Climbing Stairs

Target by Week 8

✓ Understand pointer movement ✓ Comfortable with recursion ✓ Solve Easy problems independently

✓ Phase 3 — Non-Linear DS + Interview Prep (Week 9 – 1 2)

Week 9 — Trees

Topics: - Binary Tree - DFS & BFS

Problems: - 1 0 4 — Maximum Depth of Binary Tree - 1 0 0 — Same Tree - 2 2 6
Invert Binary Tree

Week 1 0 — Binary Search

Topics: - Search space thinking

Problems: - 7 0 4 — Binary Search - 3 5 — Search Insert Position - 6 9 — Sqrt(x)

Week 1 1 — Basic Graph Thinking

Topics: - Graph representation - BFS/DFS idea

Problems: - 9 9 7 — Find the Town Judge - 1 9 7 1 — Find if Path Exists in Graph

Week 1 2 — Interview Simulation

Daily: - 1 Easy + 1 Medium problem - Time yourself (3 0 mins each)

Problems: - 1 — Two Sum - 5 3 — Maximum Subarray - 2 4 2 — Valid Anagram - 4 9 — Group Anagrams



Weekly Skill Check

Ask yourself: - Can I explain the logic aloud? - Can I code without looking? - Do I understand the solution works?

If YES → move forward. If NO → revise same topic.



Daily Rules (Very Important)

- ✓ Think 20 mins before seeing solution
 - ✓ Write code on your own
 - ✓ Debug yourself first
 - ✓ Maintain a mistake notebook
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Expected Progress

Month		Your Level
End Month	1	Strong Beginner
End Month	2	Intermediate Problem Solver
End Month	3	Interview Ready (Entry Level)



Extra Practice (Optional)

- Re-solve problems after 3 days
 - Explain solution to imaginary interviewer
 - Practice typing speed
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Final Advice

Consistency beats intelligence. Solve problems daily — even small progress counts.

You are not competing with others; you are competing with yesterday's version of yourself.

Follow this strictly for 90 days and you will see massive improvement.